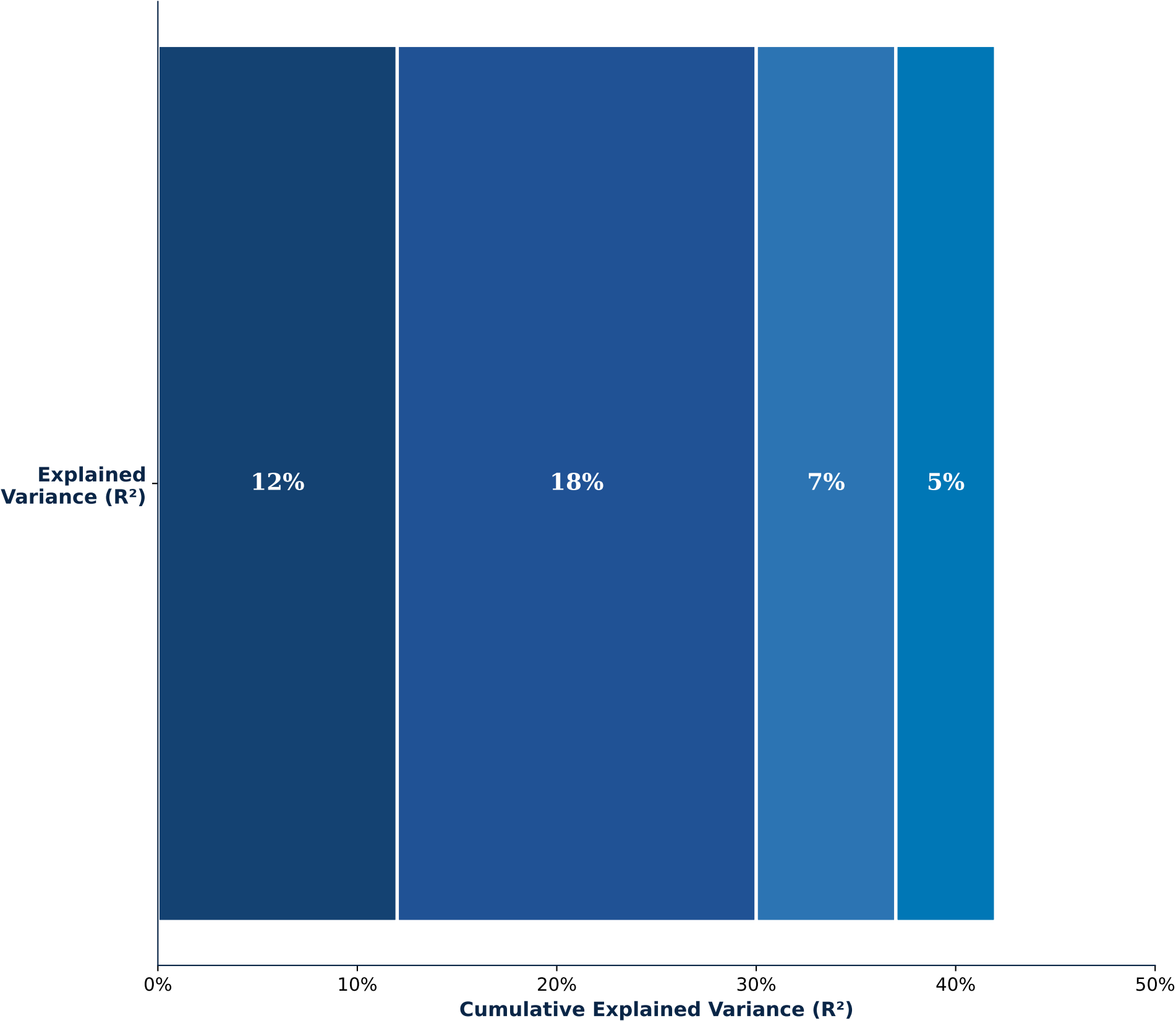


HIERARCHICAL MULTIVARIATE REGRESSION ANALYSIS

Epigenetic Age Acceleration Predictors (n = 3,847)

A. Variance Decomposition



B. Model Progression

Model 1	$R^2 = 0.12$	
Age + Sex		
↓		
Model 2	$R^2 = 0.30$	F = 287.4
+ Substance use duration	$\Delta R^2 = 0.18$	p < 0.001
↓		
Model 3	$R^2 = 0.37$	F = 94.3
+ Physiological mediators (HOMA-IR, Cortisol/ACTH, Inflammation)	$\Delta R^2 = 0.07$	p < 0.001
↓		
Model 4	$R^2 = 0.42$	F = 67.8
+ Psychological moderators (DERS, SCS-B)	$\Delta R^2 = 0.05$	p < 0.001

KEY FINDINGS

Substance use (18%) + Physiological mediators (7%) + Psychological moderators (5%)

Together explain 30% of epigenetic aging variance beyond demographics

Demographics (Age + Sex) = 12% | Physiological Mediators = 7% | Psychological Moderators: 5%

HOMA-IR = Homeostatic Model Assessment for Insulin Resistance | DERS = Difficulties in Emotion Regulation Scale | SCS-B = Self-Compassion Scale